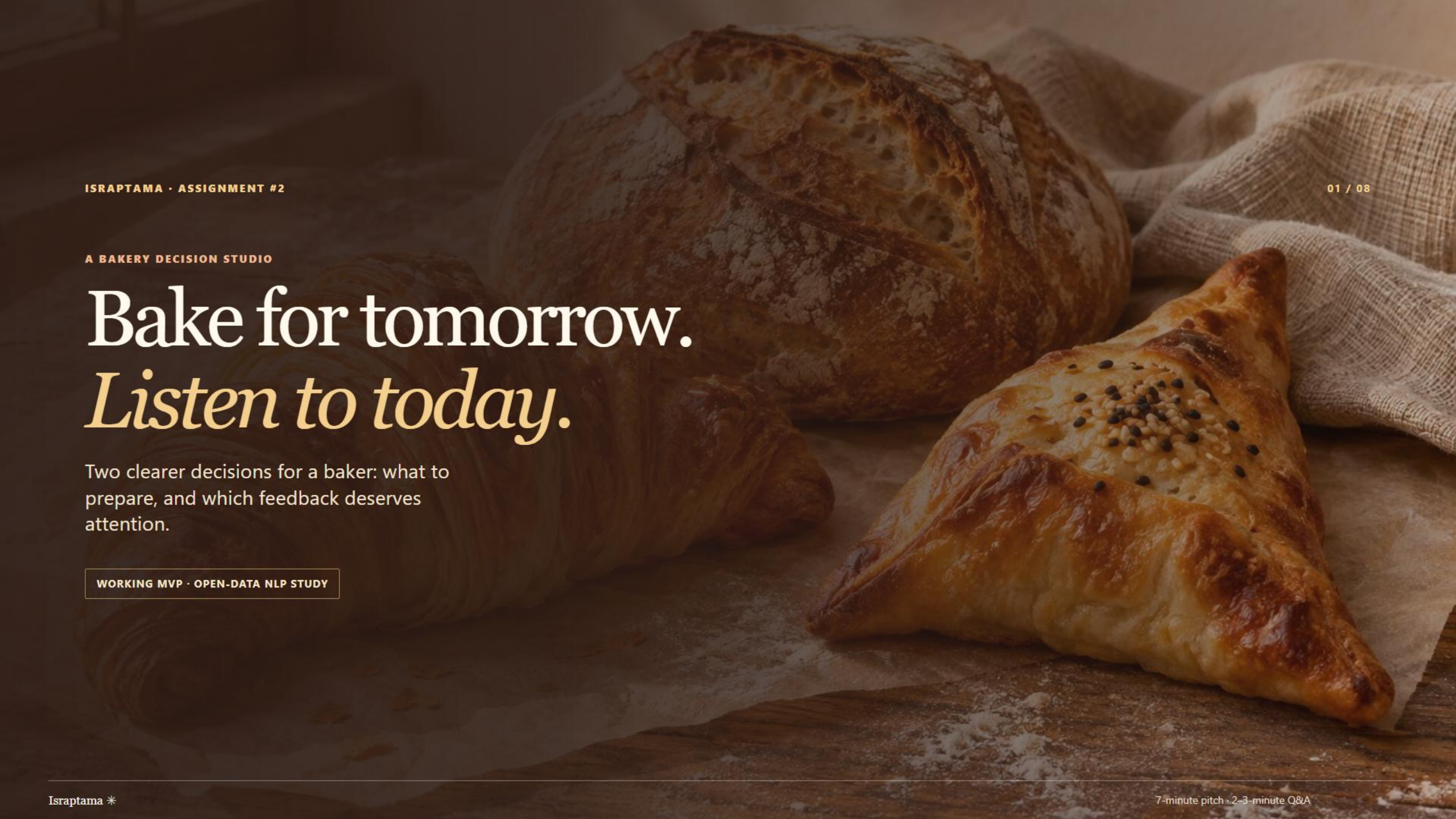
A BAKERY DECISION STUDIO

Bake for tomorrow. *Listen to today.*

Two clearer decisions for a baker: what to prepare, and which feedback deserves attention.

WORKING MVP · OPEN-DATA NLP STUDY



One bakery. *Two signals.*

01

FROM SALES HISTORY

How much should we bake?

Daily baked and sold counts become a suggested quantity for tomorrow.

02

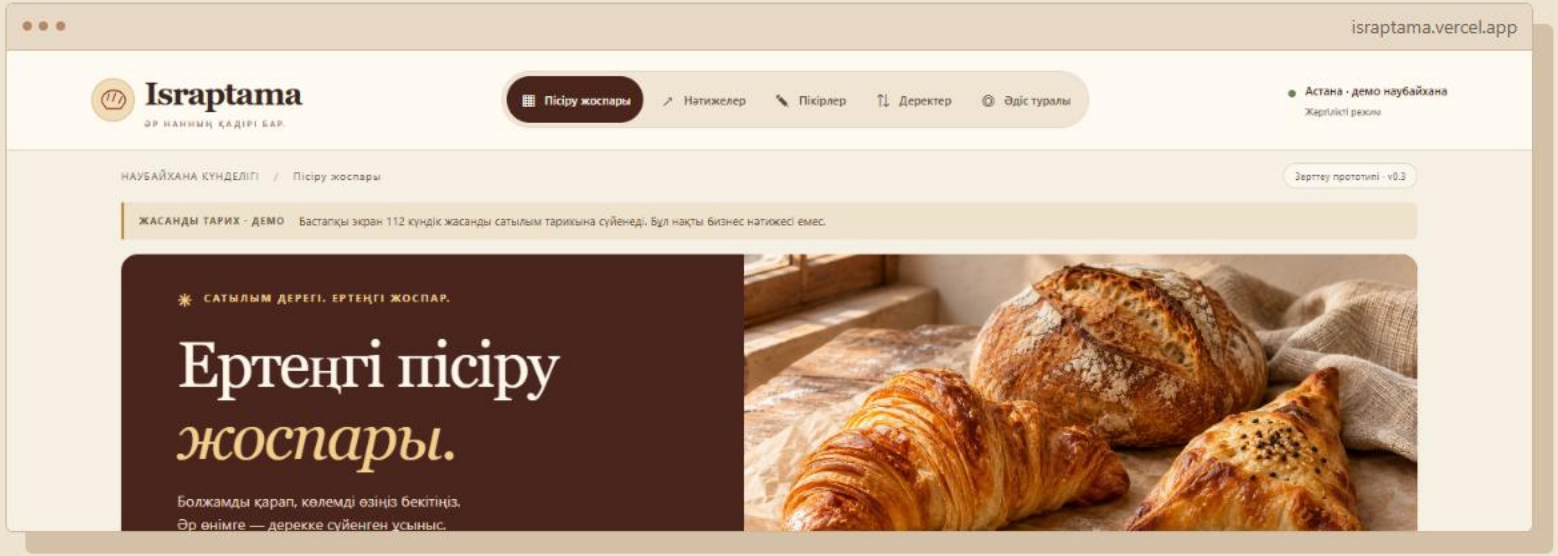
FROM WRITTEN FEEDBACK

Which comment needs a closer look?

English review text becomes a low-rating signal for a person to inspect.

Clear boundary: review classification does not change the production forecast. The planning figures shown in the demo come from synthetic bakery data.

The idea is *already on screen.*



Screenshot of the working interface. Bakery sales shown here are simulated; the NLP experiment uses separate open review data.

SHOW THE PRODUCT IN 30 SECONDS

1. Open the production plan.
2. Switch to **Пікірлер** (Reviews).
3. Classify a sample English comment.

The review view displays both model outputs and the measured comparison.

Open the live dashboard ↗

Can words flag *low-rated feedback?*

A 1–3-star rating is the target signal; 4–5 stars form the other class. This is a rating-derived proxy, not a verified complaint label.

18,182	5,984	38
UCI source comments	usable rated reviews	bakery-like recipes

Entire recipes held apart

No recipe crosses the split

TRAIN	VALIDATE	FINAL TEST
25 recipes	6 recipes	7 unseen recipes
4,011 reviews	799 reviews	1,174 reviews · 73 low-rated

Source: UCI Recipe Reviews and User Feedback ↗ · CC BY 4.0 · DOI 10.24432/C5FG95. TF-IDF fits training text only; validation chooses thresholds and the live model.

Change the text representation. *Keep the test fair.*

BASELINE

Words

Word unigram TF-IDF

↓

Balanced logistic regression

ONE PLANNED ADDITION

Words + characters

Word TF-IDF + character 3–5 grams

↓

Same logistic regression

Held fixed: recipe split, normalization, class weighting, C = 1, liblinear solver, and validation threshold rule.

More found. *More false alerts.*

LOW-RATED REVIEWS CAUGHT · OUT OF 73

36 → 48

baseline candidate

+12 found

HIGH-RATED REVIEWS FLAGGED

37 → 67

baseline candidate

+30 false alerts

Low-rating F1 **0.493 → 0.511** Average precision **0.518 → 0.558**

The 95% recipe-cluster interval for the F1 difference is **−0.066 to +0.080**: a reliable gain is not established. The live model remains the word baseline, chosen before test inspection on validation F1 (0.620155 vs 0.619355).

Prove it in a *real bakery*.

WHAT WE KNOW TODAY

- English home-recipe comments are only a proxy for bakery feedback.
- Seven test recipes leave the model gap uncertain.
- No bakery waste reduction or operational uplift has been measured.

A SMALL, HONEST PILOT

Test with one real bakery.

1. Collect opt-in feedback and local-language labels.
2. Evaluate on new products before deployment.
3. Measure triage usefulness and planning outcomes separately.

The prototype already makes its data, split, model choice, predictions, and limitations inspectable.

A WORKING IDEA, READY FOR A REAL TEST

Bake with foresight. *Listen with care.*

Give a baker one place to plan tomorrow and notice what customers are saying today.

Try the MVP ↗ Inspect the research ↗

Next ask: an opt-in bakery pilot. Business impact remains unmeasured.